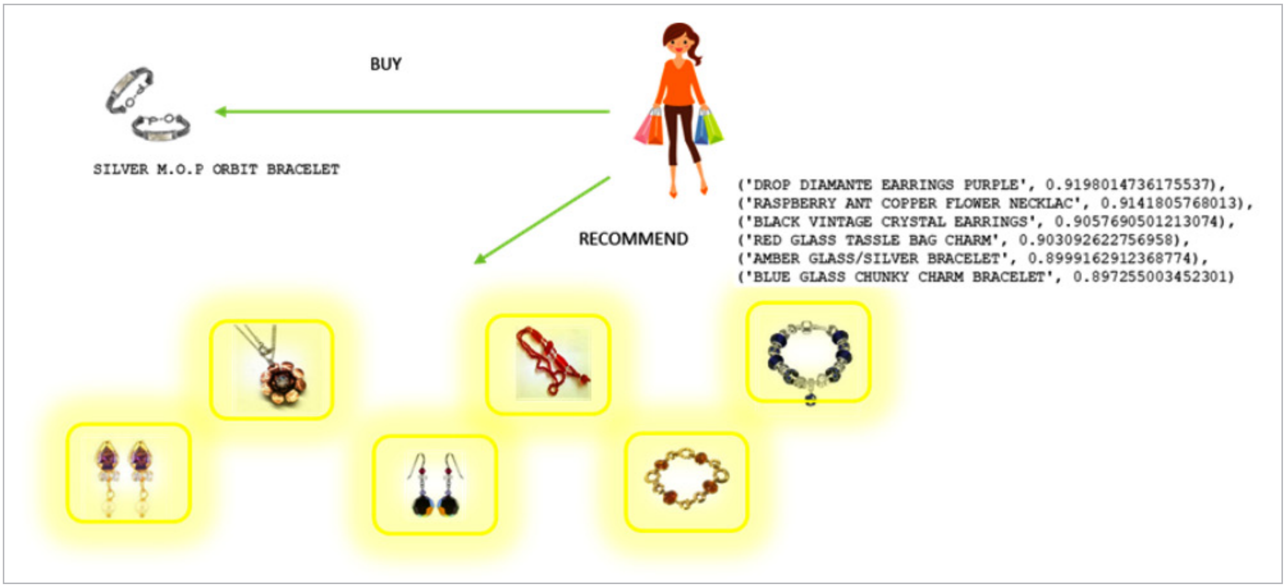

Figure 6: Recommendations

```
train_df = final_data[final_
data['CustomerID'].isin(customers_
train)]
validation_df = final_data[~final_
data['CustomerID'].isin(customers_
train)]
```

Now, for both the train and validation sets, we will build sequences of purchases made by customers in the data set.

```
purchases_train = []

for i in tqdm(customers_train): ## We
could have used tqdm(train_df)??
    temp = train_df[train_
df["CustomerID"] == i]["StockCode"].
tolist()
    purchases_train.append(temp)
purchases_val = []

for i in tqdm(validation_
df['CustomerID'].unique()):
    temp = validation_df[validation_
df["CustomerID"] == i]["StockCode"].
tolist()
    purchases_val.append(temp)

model = Word2Vec(window = 10, sg = 1,
hs = 0,
                 negative = 10, # for
negative sampling
                 alpha=0.03, min_
alpha=0.0007,
                 seed = 14)

model.build_vocab(purchases_train,
progress_per=200)

model.train(purchases_train, total_
examples = model.corpus_count,
            epochs=10, report_delay=1)
model.init_sims(replace=True)
model.save("my_word2vec_2.model")
```

Recommending products

To simply match a product's description to its ID, let's establish a product-ID and product-description dictionary.

```
products = train_df[["StockCode",
"Description"]]

# remove duplicates
products.drop_duplicates(inplace=True,
subset='StockCode', keep="last")

# create product-ID and product-
description dictionary
products_dict = products.
groupby('StockCode')['Description'].
apply(list).to_dict()

products_dict['90019A']
```

The output will be:

```
['SILVER MOP ORBIT BRACELET']
```

Let's create a function which will take a product's vector (v) as input and return the top six similar products.

```
def get_similar_item(v, n = 6):

    ms = model.similar_by_vector(v,
topn= n+1)[1:]
    my_ms = []
    for j in ms:
        pair = (products_dict[j[0]][0],
j[1])
        my_ms.append(pair)

    return my_ms
```

Let's try out our function by passing the vector of the product '90019A' ('SILVER M.O.P ORBIT BRACELET'):

```
get_similar_item(model['90019A'])
```

The output will be:

```
[('TURQUOISE GLASS TASSLE BAG CHARM',
```